

# Deep Learning and Generative AI

## Course Outline (1<sup>st</sup> Half: DL for Computer Vision)

### Week-1:

Faculty Session: Introduction, Motivation and History

- Datasets, Tasks and Challenges
- History and Course overview

TA Session: Introduction to Python and PyTorch

### Week-2:

Faculty Session: Single neuron computation

- Perceptron model and its limitations
- Deep v/s shallow network
- Universal approximation theorem

TA Session: Loss functions, Optimizers, Backpropagation and Gradient Descent

### Week-3:

Faculty Session: Feed-forward Neural Network (FNN)

- N-layer MLP + softmax
- How to train the parameters – Backpropagation
- A machine learning recipe

TA Session: Deep Learning workflow with FFN example

### Week-4:

Faculty Session: Convolutional Neural Networks (CNN)

- Convolution operation
- Convolution layer
- Pooling
- Full CNN architecture

TA Session: CNN Implementation

**Week-5:**

Faculty Session: Different CNN Architecture

- Vanishing/Exploding Gradients
- Normalization (Batch or Layer) and Regularization
- AlexNet
- VGG-Net
- Google-Net
- DenseNet

TA Session: End-to-end model for defect detection

**Week-6:**

Faculty Session: Auto-encoder and Image Segmentation

- Auto-encoder
- Regularized Auto-encoder
- Class-encoder
- Application: Image Segmentation

TA Session: --

**Week-7:**

Faculty Session: RNN and its Variants

- Sequential learning problem
- Recurrent Neural Network (RNN), BPTT
- LSTM, Deep-LSTM, Bi-directional RNN, Bi-LSTM
- Gated Recurrent Unit (GRU)
- Applications: Sentiment Analysis, Machine Translation, Image Captioning, Video Analysis

TA Session: Recap of sequence-to-sequence models and Applications